BIOLOGY

SBA: RESEARCH PROJECT

INVESTIGATIVE PRACTICAL

The Effect of Temperature on the Rate of Catalase Activity in Potato Tissue

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Catalase (potato) + hydrogen peroxide, gas collection by water displacement — six water-bath temperatures from 0°C to 100°C.

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INTRODUCTION

BACKGROUND INFORMATION

Enzymes are biological catalysts — proteins that speed up the rate of chemical reactions in living organisms without being used up themselves. Catalase is an enzyme found in the cells of most living organisms, including potato tubers and liver, where it protects cells by breaking down hydrogen peroxide (H_2O_2), a toxic by-product of metabolism, into water and oxygen gas.

$$2\ H_2O_2 \rightarrow 2\ H_2O + O_2$$

Like all enzymes, catalase is a protein with a specific three-dimensional shape, including an active site that binds to its substrate. Enzyme activity is affected by several factors — pH, substrate concentration, and, most notably, temperature. As temperature rises, molecules gain kinetic energy and collide more frequently and more successfully at the active site, increasing the rate of reaction. Beyond an optimum temperature, however, heat disrupts the bonds holding the enzyme's tertiary structure together; the active site changes shape, the substrate can no longer bind, and the enzyme is said to be *denatured*. This relationship is why we refrigerate food to slow spoilage, why a high fever is dangerous, and why hydrogen peroxide poured on a cut foams most vigorously at body temperature.

PROBLEM STATEMENT

It is not clear at what temperature catalase extracted from potato tissue is most effective at breaking down hydrogen peroxide, and how its activity changes as temperatures approach and exceed the point at which most enzymes are known to denature.

RESEARCH OBJECTIVE

To determine the effect of varying temperature on the rate of catalase activity in potato tissue, as measured by the rate of oxygen gas production when potato tissue is reacted with hydrogen peroxide.

HYPOTHESIS

> **H_1 (alternate):** The rate of catalase activity in potato tissue will increase with temperature up to an optimum at approximately human body temperature (37°C), after which the rate will decrease sharply due to denaturation of the enzyme.

> **H_0 (null):** Changes in temperature will have no significant effect on the rate of catalase activity in potato

tissue.

VARIABLES

Independent	Temperature of the reaction mixture (°C) — set using water baths at 0, 20, 37, 50, 70 and 100°C.
Dependent	Rate of reaction, measured as volume of oxygen gas (cm^3) produced per minute, via water displacement.
Controlled	Volume/concentration of H_2O_2 (20 cm^3, 20-volume ≈6%); size and mass of potato disc (1 cm diameter × 1 cm thick, same tuber); pre-incubation time (5 min); reaction time (3 min); apparatus used.

METHODOLOGY

MATERIALS / APPARATUS

Fresh potato tuber, hydrogen peroxide solution (20 volume/≈6%), cork borer (No. 4), scalpel, white tile, ruler, filter paper, forceps, boiling tubes, retort stand and clamp, delivery tube with bung, trough, inverted 50 cm^3 measuring cylinder, thermometer, stopwatch, beakers, ice, kettle/hot plate, wax pencil, distilled water.

METHOD

1. Six water baths were prepared and maintained at 0°C (ice/water), 20°C (room temperature), 37°C, 50°C, 70°C and 100°C (boiling), and the actual temperature of each was confirmed with a thermometer.
2. Potato discs of equal size (1 cm diameter, 1 cm thick) were cut using a cork borer and scalpel on a white tile, and blotted dry with filter paper.
3. 20 cm^3 of hydrogen peroxide was measured into a boiling tube and placed into the first water bath.
4. The hydrogen peroxide was left to equilibrate to the water bath temperature for 5 minutes.
5. A potato disc was added to the boiling tube, which was immediately sealed with a bung fitted with a delivery tube directed into an inverted, water-filled measuring cylinder standing in a trough of water.
6. The stopwatch was started immediately, and the volume of oxygen gas collected was read and recorded every 30 seconds for 3 minutes.
7. Steps 3–6 were repeated twice more at the same temperature (three trials), then the whole procedure was repeated at each of the remaining five temperatures.
8. The average volume of oxygen produced after 3 minutes was calculated for each temperature, and the rate of reaction (cm^3/min) found by dividing the average volume by the reaction time.

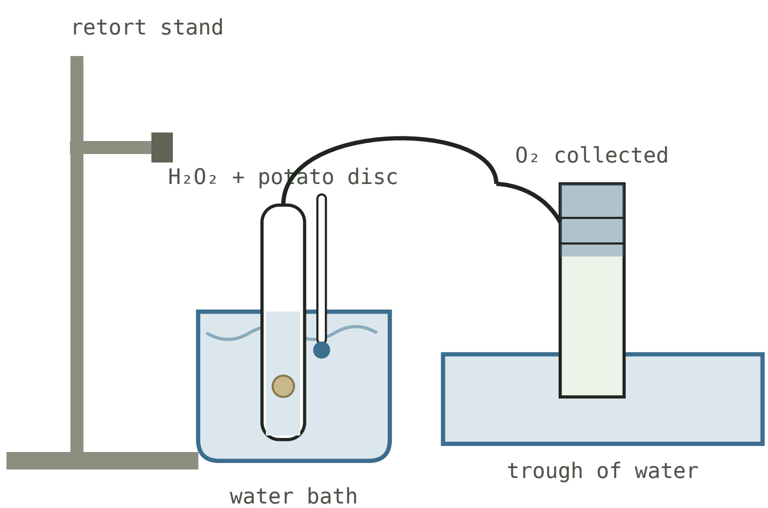

Fig. A — Gas collection by water displacement: a potato disc reacts with hydrogen peroxide in a temperature-controlled water bath; oxygen produced is collected in an inverted measuring cylinder.

SAMPLE CALCULATION

Temperature = 37°C · Trial 1 = 23.0 cm^3 · Trial 2 = 24.5 cm^3 · Trial 3 = 22.0 cm^3

Average volume of O_2 = (23.0 + 24.5 + 22.0) ÷ 3 = 23.2 cm^3

Rate of reaction = Volume of O_2 ÷ Time = 23.2 cm^3 ÷ 3 min = **7.7 cm^3/min**

> **Limitation noted during testing:** the water bath temperature may not have been identical to the temperature inside the reaction tube itself, meaning the recorded temperature may not fully reflect actual reaction conditions.

PRESENTATION OF DATA

OBSERVATION / RESULTS

TABLE 1 — Volume of oxygen produced and rate of catalase activity at different temperatures

TEMP (°C)	TRIAL 1 (CM^3)	TRIAL 2 (CM^3)	TRIAL 3 (CM^3)	AVERAGE (CM^3)	RATE (CM^3/MIN)
0	1.5	1.0	1.5	1.3	0.4
20	8.0	7.5	8.5	8.0	2.7
37	**23.0**	**24.5**	**22.0**	**23.2**	**7.7**
50	11.0	10.0	10.5	10.5	3.5
70	3.0	2.5	2.0	2.5	0.8
100	0.5	0.0	0.5	0.3	0.1

LINE GRAPH

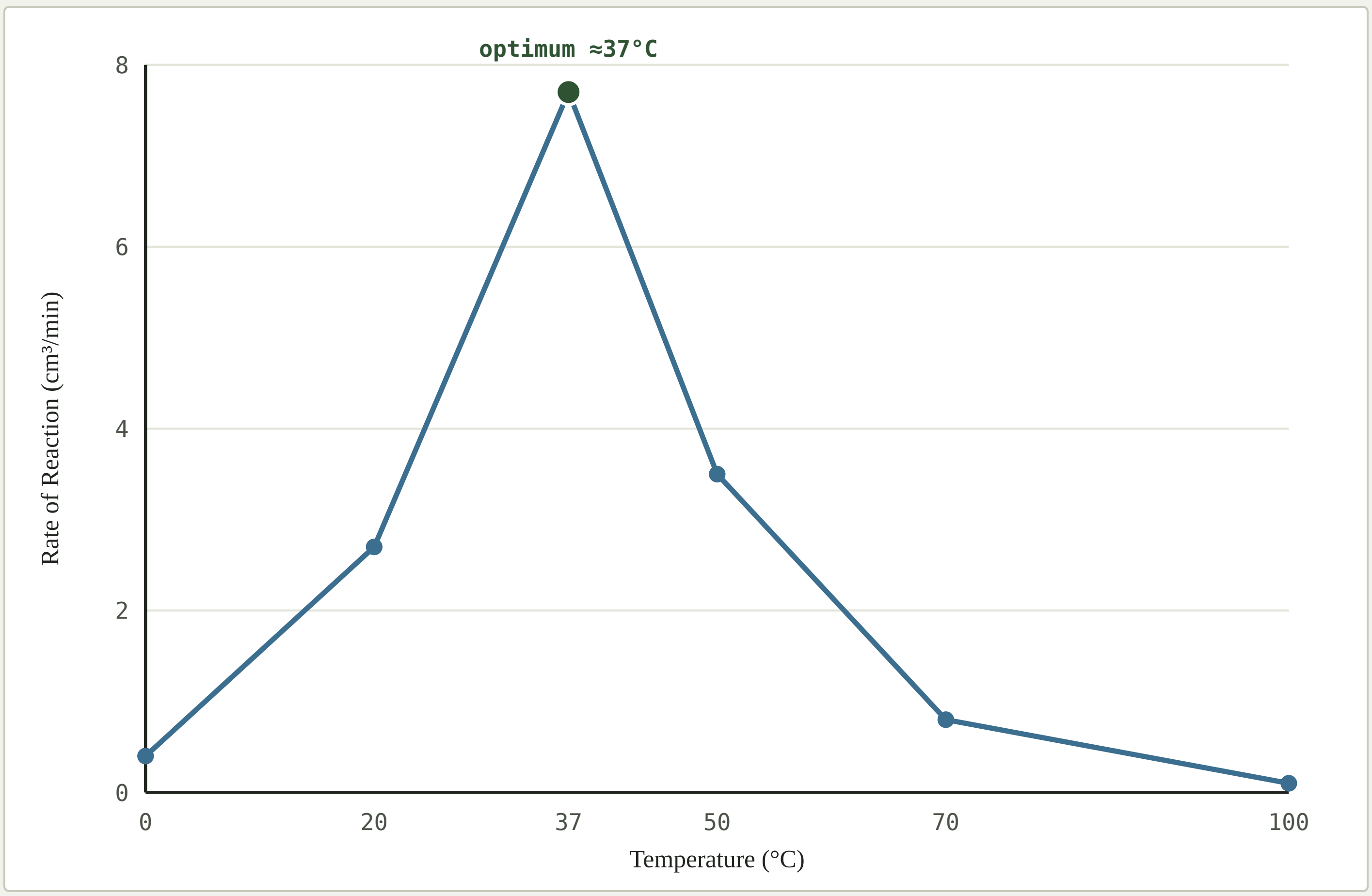

Fig. 1 – Rate of catalase activity in potato tissue at six temperatures, showing a peak at approximately 37°C followed by a sharp decline consistent with enzyme denaturation.

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ANALYSIS AND INTERPRETATION OF DATA

The results show a clear relationship between temperature and the rate of catalase activity in potato tissue. At 0°C, the rate of reaction was very low (0.4 cm^3/min) because the low kinetic energy of the molecules resulted in few successful collisions between the enzyme's active site and the hydrogen peroxide substrate; the enzyme itself was not damaged at this temperature, only inactive.

As temperature increased to 20°C and then 37°C, the rate of reaction rose sharply, reaching a peak of 7.7 cm^3/min at 37°C. This is consistent with the kinetic theory of matter: as temperature rises, molecules move faster and collide more frequently and more energetically, increasing the number of successful enzyme–substrate collisions per second. 37°C represents the optimum temperature for this enzyme, close to normal human body temperature, at which the active site is in its ideal conformation for binding hydrogen peroxide.

Beyond 37°C, the rate of reaction fell steadily — 3.5 cm^3/min at 50°C, 0.8 cm^3/min at 70°C, and almost zero (0.1 cm^3/min) at 100°C. This decline reflects denaturation: excess heat disrupts the hydrogen and ionic bonds maintaining the enzyme's tertiary structure, permanently changing the shape of the active site so hydrogen peroxide can no longer bind effectively. By 100°C, the enzyme was almost completely denatured and catalase activity had virtually ceased.

These findings mirror real-world phenomena: refrigeration slows food spoilage by reducing the rate of enzyme-catalysed reactions; a fever, if too high, risks denaturing essential human enzymes; and hydrogen peroxide is most effective as a wound antiseptic at body temperature, where catalase in the blood reacts with it most readily.

CONCLUSION

The results support the alternate hypothesis: the rate of catalase activity in potato tissue increased with temperature up to an optimum of approximately 37°C, after which it decreased sharply due to denaturation of the enzyme. This confirms that catalase, like other enzymes, is a temperature-sensitive protein whose activity is maximised within a narrow optimum range and destroyed by excessive heat.

LIMITATIONS

- The water bath temperature may not have exactly matched the temperature inside the reaction tube, especially in the first seconds after the potato disc was added.
- Manual timing and reading of gas volumes introduces human/reaction-time error.
- Natural variation in catalase concentration may exist between different potato tubers, and even within the same tuber.
- Only three trials were carried out at each temperature, limiting the statistical reliability of the averages.
- The hydrogen peroxide solution may have gradually decomposed with exposure to light and air over the course of testing, slightly reducing its concentration in later trials.

RECOMMENDATIONS

- Use a gas-pressure sensor and datalogger for continuous, more precise readings of oxygen production instead of manual water displacement.
- Use a standardised, homogenised potato extract of known concentration rather than solid discs, to reduce variation in catalase concentration between samples.
- Test more temperature intervals close to the suspected optimum (e.g., 30°C, 35°C, 40°C, 45°C) to more precisely pinpoint the true optimum.
- Measure the actual temperature inside the reaction tube directly, rather than relying solely on the water bath temperature.
- Increase the number of trials at each temperature to improve the reliability of the averages obtained.

REFLECTION

Carrying out this investigation helped me understand, in a hands-on way, how temperature affects enzyme activity and why enzymes have an optimum temperature rather than simply working faster the hotter it gets. Seeing the reaction almost stop completely at 100°C made the concept of denaturation far more concrete than reading about it in a textbook.

This experiment also helped me appreciate why temperature control matters in everyday life — from refrigerating food to slow spoilage, to why a high fever can be dangerous, to why industrial enzymes must be used within specific temperature ranges.

If I were to extend this investigation, I would like to compare catalase activity in potato tissue with that in liver tissue, which contains a much higher concentration of catalase, to see whether the same optimum temperature applies. I would also like to investigate the effect of pH on catalase activity using a similar experimental design.

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APPENDIX

APPENDIX I — DIAGRAM OF APPARATUS

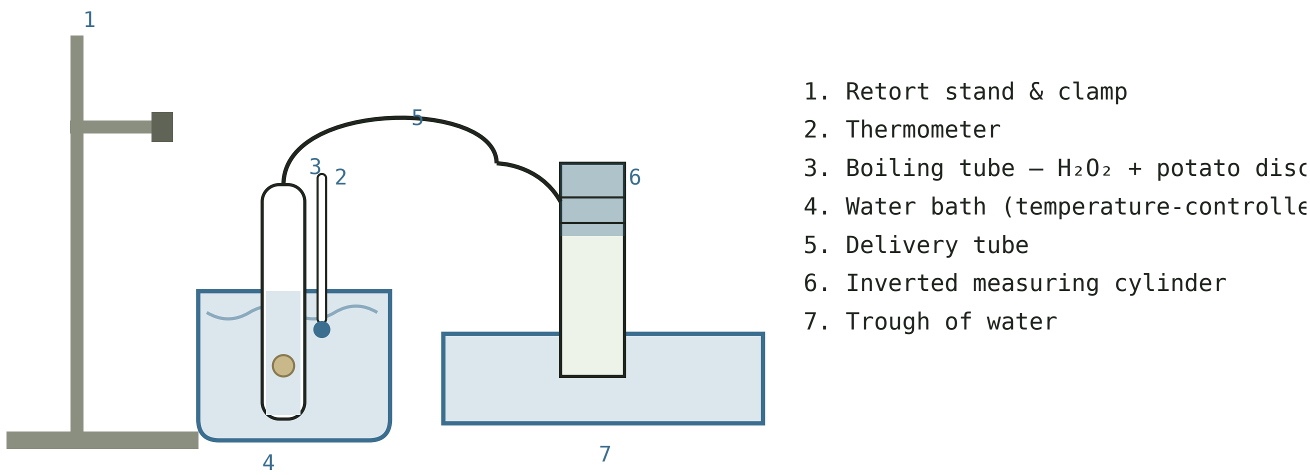

Fig. A1 – Labelled diagram of the gas-collection-by-water-displacement apparatus used at each of the six test temperatures.

APPENDIX II — RAW DATA

See Table 1 in the Presentation of Data section for the full set of raw readings (Trials 1–3) at each of the six temperatures tested.